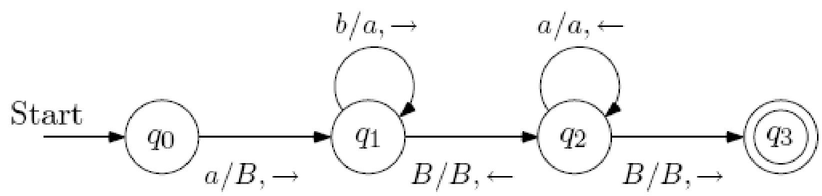
CSE 309 Theory of Automata, Spring 2026
Exercises 5

1. **Interpretation of Turing Machines** Consider a Turing machine M_1 that processes strings over the input alphabet $\Sigma = \{a, b\}$ and whose transition diagram is as follows. As usual, "B" is the blank symbol for this Turing machine.



   (a) Is it true that $abq_1bBb \vdash_{M_1} abaq_1Bb$?

   (b) Is it true that $abq_1bBb \vdash^*_{M_1} abaq_1Bb$

   (c) Trace the execution of this Turing machine on each of the input strings a, b, ab and abb.

   (d) What is the language of this machine?

   (e) Does M_1 decide this language?

   (f) Suppose w is a string that is accepted by the above Turing machine. Describe the contents of the tape when M_1 enters the final state q_3 (after being run on the input w).

2. **Design of Turing Machines** Let $L_2 = \{0^i\#0^j\#0^k \mid i+j = k\}$. One possible Turing machine algorithm that decides this language is as follows. On input string w,

   Step 1: Check whether $w \in L(0^*\#0^*\#0^*)$, and reject w if this is not the case.

   Step 2: Match all 0's in the first group with one zero in the last group. If there are insufficient zeroes in the last group reject.

   Step 3: Match all 0's in the middle group with the remaining zeroes in the last group. If there are insufficient zeroes remaining in the last group, reject. If there are unmatched zeroes remaining in the last group after this step, reject.

   (a) Give the transition diagram of a subroutine Turing machine that implements the first step in the above algorithm. Let the start state for this subroutine be r_1^0. If the input string w is of the required form, the subroutine should terminate in state r_1^1. If the input string is not of the required form, the subroutine should terminate in state r_1^f. When the subroutine is in state r_1^1 the tape head should be pointing to the beginning of the input string again.

(b) Give a transition diagram of a subroutine Turing machine that implements the second step in the above algorithm. Let the start state for this subroutine be r_2^0, let the state that indicates that the input string passes this step be r_2^1 and let the rejection state be r_2^f. Use the programming technique of multiple tracks on the tape to match zeroes. When the subroutine is in state r_2^1 the tape head should be pointing to the first zero in the second set of zeroes.

(c) Give a transition diagram of a subroutine Turing machine that implements the third step in the above algorithm. Let the start state for this subroutine be r_3^0, let the state that indicates that the input string passes this step be r_3^1 and let the rejection state be r_3^f.

(d) Create a Turing machine that uses these three subroutine Turing machines that decides the language L_2.